

Theory of ARIMA Time Series Models

Opening Question

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Stationarity tells us that we can do so for a time series (please explain.....).

Introduction to AR Models

To appreciate time series modelling we will first begin with:

- ① An AR (Autoregressive) model
- ② An Example of a situation we would like to model

AR Example

Background: Kojo was hired on Monday. During his first week, we observed some variability in his arrival time: on some days, he arrived before the scheduled start time of 8:00 a.m., while on other days, he arrived after 8:00 a.m.

Task at hand: We want to develop a model that predicts Kojo's *deviation* from the scheduled start time so that we can quantify and understand the factors associated with variations in his arrival time.

Kojo's Arrival-Time Deviations

Day	Deviation (minutes)
Day 1	-7
Day 2	+4
Day 3	-2
Day 4	+9
Day 5	-4

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Day 5	-4

What is the average deviation from the sample we have?

Kojo's Arrival-Time Deviations

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In time series, past observations can provide information about future observations. Therefore, we can use Kojo's previous deviation from the start time to predict his deviation on the current day.

Kojo's Arrival-Time Deviations

$$deviation_t = \phi deviation_{t-1}$$

Here, ϕ measures the **strength and direction of the relationship** between the current deviation and the previous deviation.

- 1 If $\phi = 0$, the previous deviation provides no information about the current deviation.

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- 2 If $0 < \phi < 1$, deviations tend to persist in the same direction. A positive deviation (late arrival) tends to be followed by another positive deviation (late arrival) and vice versa.

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- 2 If $0 < \phi < 1$, deviations tend to persist in the same direction. A positive deviation (late arrival) tends to be followed by another positive deviation (late arrival) and vice versa.
- 3 If $-1 < \phi < 0$, deviations tend to reverse direction. A positive deviation (late arrival) tends to be followed by a negative deviation (early arrival) and vice versa.

Kojo's Arrival-Time Deviations

What direction is ϕ likely to be, given the table below:

$$deviation_t = \phi deviation_{t-1}$$

Day	Deviation (minutes)
Day 1	-7
Day 2	+4
Day 3	-2
Day 4	+9
Day 5	-4

Kojo's Arrival-Time Deviations

Assuming $\phi = -0.8$ what is our prediction for each day?

$$deviation_t = -0.8deviation_{t-1}$$

Day	Deviation (minutes)	Prediction
Day 1	-7	-
Day 2	+4	?
Day 3	-2	?
Day 4	+9	?
Day 5	-4	?

Kojo's Arrival-Time Deviations

Assuming $\phi = -0.8$ what is our prediction for each day?

$$deviation_t = -0.8deviation_{t-1}$$

Day	Deviation (minutes)	Prediction
Day 1	-7	-
Day 2	+4	5.6
Day 3	-2	-3.2
Day 4	+9	1.6
Day 5	-4	-7.2

Kojo's Arrival-Time Deviations

Fill the remaining columns based on your findings:

Day	Deviation	Prediction	Pred Error	Absolute Pred Error
Day 1	-7	-	-	-
Day 2	+4	5.6	-1.6	1.6
Day 3	-2	-3.2	?	?
Day 4	+9	1.6	?	?
Day 5	-4	-7.2	?	?
Average	0	-0.8	?	?

Kojo's Arrival-Time Deviations

Fill the remaining columns based on your findings:

Day	Deviation	Prediction	Pred Error	Absolute Pred Error
Day 1	-7	-	-	-
Day 2	+4	5.6	-1.6	1.6
Day 3	-2	-3.2	1.2	1.2
Day 4	+9	1.6	7.4	7.4
Day 5	-4	-7.2	3.2	3.2
Average	0	-0.8	2.55	3.35

Kojo's Arrival-Time Deviations

Fill the remaining columns based on your findings:

Day	Deviation	Prediction	Pred Error	Absolute Pred Error
Day 1	-7	-	-	-
Day 2	+4	5.6	-1.6	1.6
Day 3	-2	-3.2	1.2	1.2
Day 4	+9	1.6	7.4	7.4
Day 5	-4	-7.2	3.2	3.2
Average	0	-0.8	2.55	3.35

Above makes it clear that the true AR model representing Kojo's Arrivals are as follows:

$$deviation_t = -0.8deviation_{t-1} + \epsilon$$

Exercise 1: Ama's Arrival-Time Deviations

- You remember that Ama (another intern) also has variable arrival times
- Estimate an AR1 model for Ama (assume your own parameter, $deviation_t = \phi deviation_{t-1}$) and fill the table below
- Person with the lowest forecast error gets a prize

Day	Deviation	Prediction	Pred Error	Absolute Pred Error
Day 1	6	-	-	-
Day 2	-3	?	?	?
Day 3	1	?	?	?
Day 4	-8	?	?	?
Day 5	-4	?	?	?
Average	0	?	?	?

Back to Kojo

Introducing the ARMA Model. Recall the AR Model and the resulting table

$$deviation_t = -0.8deviation_{t-1} + \epsilon$$

Day	Deviation	AR Prediction	AR Pred Error	ARMA Prediction
Day 1	-7	-	-	-
Day 2	+4	5.6	-1.6	?
Day 3	-2	-3.2	1.2	1.2
Day 4	+9	1.6	7.4	7.4
Day 5	-4	-7.2	3.2	3.2
Average	0	-0.8	2.55	3.35

We are going to create something similar but now with an ARMA model

$$deviation_t = -0.8deviation_{t-1} + \beta\epsilon_{t-1} + \epsilon$$

What is different?

Introducing the ARMA Model (where $\beta = 0.5$ and $\epsilon_0 = 0$)

$$deviation_t = -0.8deviation_{t-1} + 0.5\epsilon_{t-1} + \epsilon$$

Day	Deviation	ARMA Error	ARMA Prediction
Day 1	-7	0	-
Day 2	+4	?	5.6
Day 3	-2	-	-
Day 4	+9	-	-
Day 5	-4	-	-
Average	0	-	-

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Day 1	-7	0	-
Day 2	+4	-1.6	5.6
Day 3	-2	-	-
Day 4	+9	-	-
Day 5	-4	-	-
Average	0	-	-

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Day	Deviation	ARMA Error	ARMA Prediction
Day 1	-7	0	-
Day 2	+4	-1.6	5.6
Day 3	-2	?	-4
Day 4	+9	-	-
Day 5	-4	-	-
Average	0	-	-

Introducing the ARMA Model (where $\beta = 0.5$ and $\epsilon_0 = 0$)

$$deviation_t = -0.8deviation_{t-1} + 0.5\epsilon_{t-1} + \epsilon$$

Day	Deviation	ARMA Error	ARMA Prediction
Day 1	-7	0	-
Day 2	+4	-1.6	5.6
Day 3	-2	2	-4
Day 4	+9	-	-
Day 5	-4	-	-
Average	0	-	-

With this understanding kindly fill the entire table!

Introducing the ARMA Model (where $\beta = 0.5$ and $\epsilon_0 = 0$)

$$deviation_t = -0.8deviation_{t-1} + 0.5\epsilon_{t-1} + \epsilon$$

Day	Deviation	ARMA Error	ARMA Prediction
Day 1	-7	0	-
Day 2	+4	-1.6	5.6
Day 3	-2	2	-4
Day 4	+9	6.4	2.6
Day 5	-4	0	-4
Average	0	1.36	0.05

AR Compared to ARMA Model

$$deviation_t = -0.8deviation_{t-1} + \epsilon$$

$$deviation_t = -0.8deviation_{t-1} + 0.5\epsilon_{t-1} + \epsilon$$

Day	Deviation	AR Pred.	ARMA Pred.	AR Pred. Error	ARMA Pred. Error
Day 1	-7	-	-	-	-
Day 2	+4	5.6	5.6	-1.6	-1.6
Day 3	-2	-3.2	-4	1.2	2.0
Day 4	+9	1.6	2.6	7.4	6.4
Day 5	-4	-7.2	-4	3.2	0.0
Average	0	-0.8	0.05	2.55	1.36

Which one do you prefer?

Exercise 2

Populate this table for AMA the intern

$$deviation_t = -0.8deviation_{t-1} + 0.5\epsilon_{t-1} + \epsilon$$

Day	Deviation	ARMA Error	ARMA Prediction
Day 1	6	0	-
Day 2	-3	?	-
Day 3	1	-	-
Day 4	-8	-	-
Day 5	-4	-	-
Average	0	-	-

Exercise 2

Populate this table for AMA the intern

$$deviation_t = -0.8deviation_{t-1} + 0.5\epsilon_{t-1} + \epsilon$$

Day	Deviation	ARMA Error	ARMA Prediction
Day 1	6	0	-
Day 2	-3	-7.8	4.8
Day 3	1	2.5	-1.5
Day 4	-8.5	-8.5	0.5
Day 5	-6.2	- 6.2	2.2
Average	0	4.0	1.5

Variations of ARMA

- 1 What this example has shown is what is called an $\text{ARMA}(p,q)$ model where p is order 1 for the AR component and q is order 1 for the MA component.

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- 2 The ARMA model can in general have different orders for the AR and MA components. For example $\text{ARMA}(3,1)$; $\text{ARMA}(4,2)$; $\text{ARMA}(5,2)$.

- 1 What this example has shown is what is called an $\text{ARMA}(p,q)$ model where p is order 1 for the AR component and q is order 1 for the MA component.
- 2 The ARMA model can in general have different orders for the AR and MA components. For example $\text{ARMA}(3,1)$; $\text{ARMA}(4,2)$; $\text{ARMA}(5,2)$.
- 3 In the general case what we have learnt for the simple ARMA case would still apply.

ARMA(1,1)

The ARMA(1,1) model is given by:

$$Y_t = c + \phi Y_{t-1} + \theta \epsilon_{t-1} + \epsilon_t$$

where:

- ϕY_{t-1} captures the autoregressive component
- $\theta \epsilon_{t-1}$ captures the moving-average component
- ϵ_t is the current random error
- c is the constant

The General ARMA Model

An ARMA(p, q) model combines autoregressive and moving-average components:

$$Y_t = c + \sum_{i=1}^p \phi_i Y_{t-i} + \sum_{j=1}^q \theta_j \epsilon_{t-j} + \epsilon_t$$

where:

- p = order of the autoregressive (AR) component
- q = order of the moving-average (MA) component
- ϕ_i = AR coefficients
- θ_j = MA coefficients
- ϵ_t = random error (innovation) at time t
- c = constant term which determines the stationary (long run) value of the model. In our example it was 0.

ARIMA Model

An ARMA model applied to a **stationary** differenced series is called an **ARIMA** model. The number of differences determines the **integration order** d :

- First difference $\rightarrow d = 1$
- Second difference $\rightarrow d = 2$
- n th difference $\rightarrow d = n$

Thus, an $\text{ARIMA}(p, d, q)$ model combines:

$$\text{AR}(p) + \text{Differencing}(d) + \text{MA}(q)$$

The ARIMAX Model

An ARIMAX model extends ARIMA by incorporating **exogenous variables** as predictors:

$$Y_t = c + \sum_{i=1}^p \phi_i Y_{t-i} + \sum_{j=1}^q \theta_j \epsilon_{t-j} + \sum_{k=1}^K \beta_k X_{k,t} + \epsilon_t$$

where:

- p = AR order
- d = order of differencing
- q = MA order
- $X_{k,t}$ = exogenous (external) variables
- β_k = effects of the exogenous variables
- c = constant term which determines the stationary (long run) value of the model. In our example it was 0.

$$\text{ARIMA}(p, d, q) + \text{External Predictors } X_t = \text{ARIMAX}(p, d, q)$$

Test your Understanding so FAR

- 1 In your own words why is stationarity important?
- 2 What is an AR model?
- 3 What is an ARMA Model?
- 4 If one indicates an ARMA(p,q) what does that mean?
- 5 If one indicates an ARIMAX (p,d,q) what does that mean?

$SARMA(1, 1, 1, 1)_{12}$

The $SARMA(1, 1, 1, 1)_{12}$ model is given by:

$$Y_t = c + \phi Y_{t-1} + \gamma Y_{t-12} + \theta \epsilon_{t-1} + \beta \epsilon_{t-12} + \epsilon_t$$

where:

- ϕY_{t-1} captures the autoregressive component
- γY_{t-12} captures the seasonal autoregressive component
- $\theta \epsilon_{t-1}$ captures the moving-average error component
- $\beta \epsilon_{t-12}$ captures the seasonal moving-average error component
- ϵ_t is the current random error
- c is the constant

Thank You!